

FACULTY OF SCIENCE · WORKING SESSION

How will Science degrees hold up to AI?

An independent durability read of the faculty's eleven programs — brought as a **discussion, not a verdict**. The point of today is to find where this is useful, and where it breaks.

WHAT IT MEASURES

A durability score, not another ranking

DFVA — Degree Future-Viability Assessment — scores a program on **eleven dimensions** (ten core + an Irreplaceability premium), each 0–3, to a **/36** total on a four-band scale. It's an **independent, evidence-based read that complements accreditation** — it does not replace it.

RESILIENT

28–36 · AI augments graduates

MODERATE

20–27 · pivot available

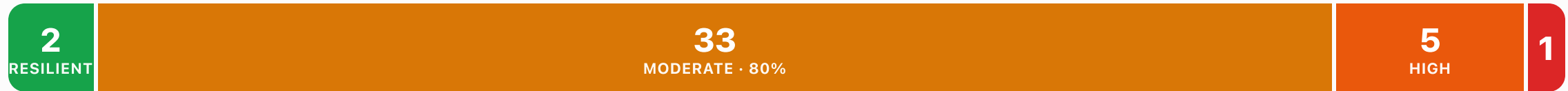
HIGH RISK

12–19 · value eroding

CRITICAL

0–11 · rapid displacement

Almost everything sits in the middle



AI Literacy is the near-universal gap

Average **1.22 / 3** across all 41 programs. Only **4** score the full 3. It is the single dimension holding the portfolio out of RESILIENT.

The threshold is hard for one reason

Programs are strong on domain and technical depth. What separates MODERATE from RESILIENT is almost always **AI literacy and outcome evidence**.

Strong programs, one narrow gap

11

programs assessed

24.0_{/36}

average — all MODERATE, zero
HIGH/CRITICAL

3.0_{/3}

Domain Depth — a perfect score

1.1_{/3}

AI Literacy — the weak point

These are **not failing degrees**. The faculty is structurally sound — deep, technical, research-rigorous. Two dimensions stand between **strong** and **resilient**: **AI Literacy (1.1)** and **Outcome Evidence (1.1)**.

THE ELEVEN, BY SCORE

All MODERATE — the spread is 21 to 26

Master of Climate Science		26	Master of Science (Physics)		25
Master of Science (BioSciences)		25	Master of Science (Chemistry)		25
Master of Biotechnology		24	Master of Science (Earth Sciences)		24
Master of Data Science		24	Master of Science (Bioinformatics)		24
Bachelor of Science		23	Master of Environmental Science		23
Master of Food Science		21			

WHERE THE POINTS ARE

The twin gap — and one of them is nearly free

10_{/11}

AI Literacy at 1/3

Ten of eleven programs score 1 on AI literacy and governance.
This is a genuine curriculum addition — but a well-scoped one.

1_{/3}

Outcome Evidence at 1/3

Low **despite** real QILT employment + salary data already in the system. This is a **data-folding gap, not an outcomes gap** — an easy lift.

Closing both is what moves programs toward RESILIENT

The gap is addressable by curriculum

Master of Climate Science

26_{/36}

Two points from RESILIENT. Strong on systems, technical and domain — one AI-methods unit would cross the line.

Master of Data Science — the exemplar

2_{/3}

The **only** Science program above the AI-literacy floor. Where computing is co-taught, the gap closes — proof this is a **curriculum** fix, not a structural one.

Two moves, run faculty-wide

1 • Embed AI literacy

A shared AI-literacy + governance core that every Science stream can draw on (D5 → 3).

2 • Fold in outcome data

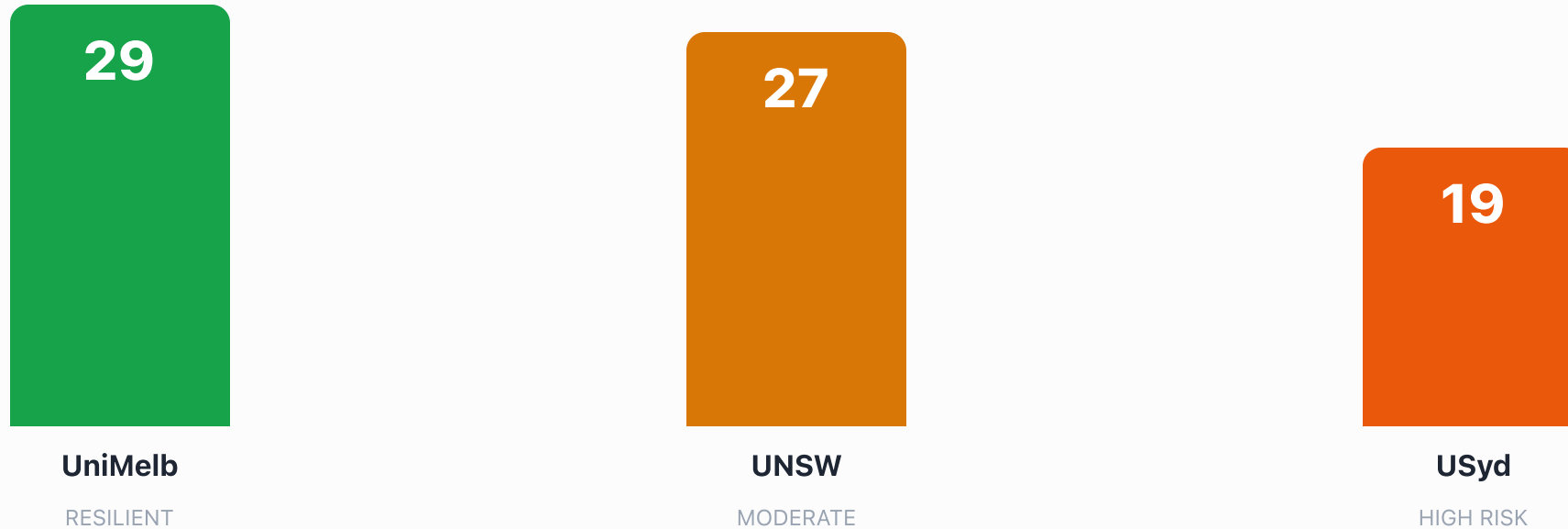
Score the QILT employment evidence already in the system into the assessment (D10 → 2+).

4₁₁

reach RESILIENT on an illustrative two-move pass. The interventions are **shared** — so a coordinated faculty initiative beats eleven separate fixes.

HOW MELBOURNE COMPARES · SAME RUBRIC, PEER PROGRAMS

Benchmarked against the Go8



Master of Computer Science, like-for-like. Worth naming the caveat: **public curriculum transparency correlates with higher scores** — open handbooks are easier to assess well.

Source: DFVA Go8 comparison · illustrative of method, not a league table

What COMPASS is — and what it isn't

IT IS

- ✓ An external durability signal for a program
- ✓ A curriculum diagnosis — where to intervene, and the likely effect
- ✓ Evidence-based and comparable across programs

IT IS NOT

- ✗ Accreditation, or a substitute for it
- ✗ A ranking or a league table
- ✗ A replacement for academic judgement

THE REAL PURPOSE OF TODAY

What I want to learn from you

01 When a Science program is reviewed, does current labour-market evidence actually reach the table?

Walk me through the last review you sat in — what was in the room?

02 Would an independent durability signal ever be welcome here — or would it be intolerable?

What would make an outside number on your degrees acceptable, or not?

03 Who would actually act on this — and is there budget for curriculum intelligence?

Distinct from what careers / IR already spend.

I'm here to find the reasons this shouldn't exist — please try to talk me out of it



THANK YOU

This is a conversation, not a pitch. Try to break it.

Take the one-pager and the dashboard with you. If a single program comes to mind that this would help — or expose — that's the next conversation.